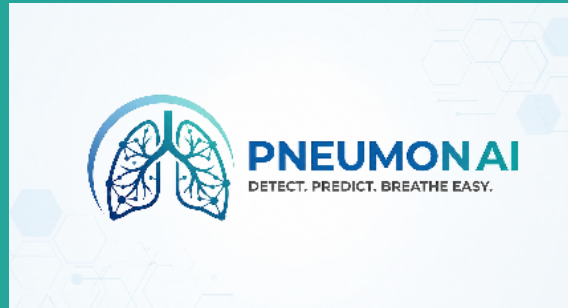




Pneumon AI Research Presentation

Ashok, Bharat, Matthew, Rachel, Saanvi, Shankar, Sharan
Team Lead: Natasha

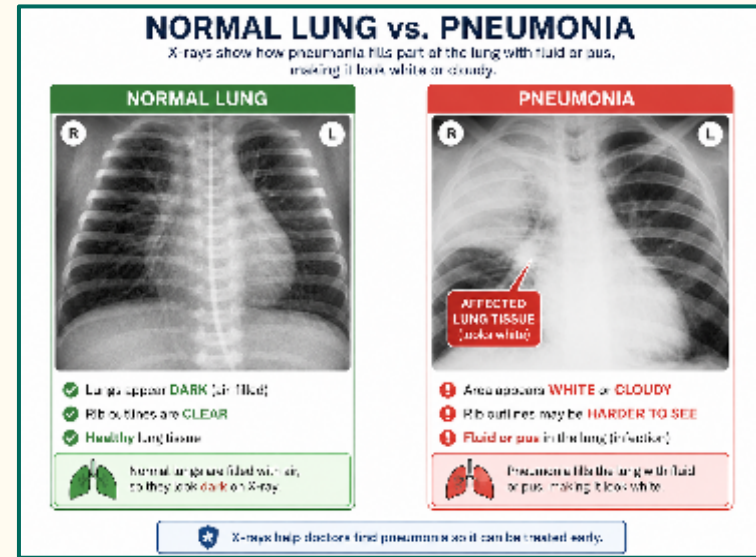
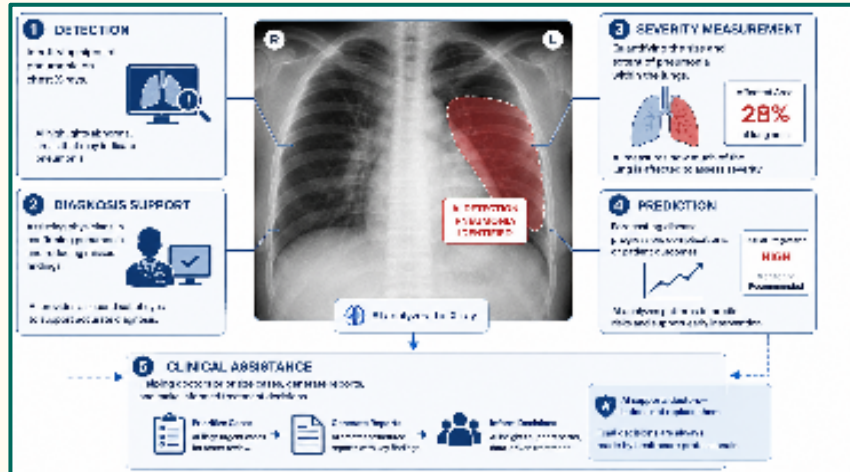
1. The Clinical Problem

—

What is Pneumonia?

- Lung infection that affects millions of people worldwide each year.
- Leading Causes: bacteria and viruses
- Global Impact: 2.5 million deaths annually (National Library of Medicine)

How Can AI Help?



- Detection
- Diagnosis Support
- Severity Measurement
- Prediction
- Clinical Assistance

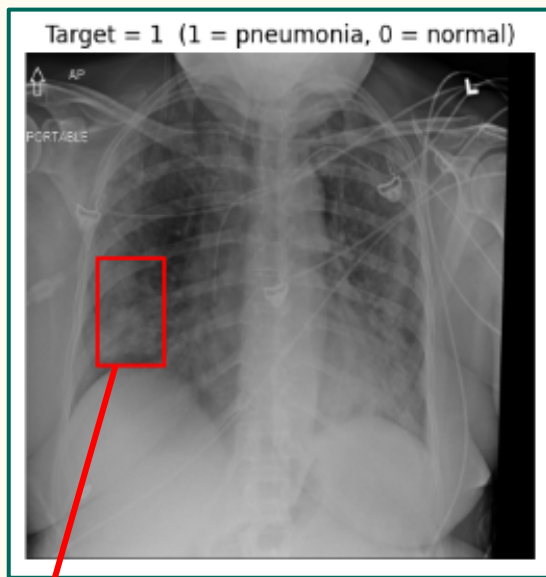
Project Goal

Build ML models to reliably detect Pneumonia and its location using chest x-rays while **mitigating false negatives.**

Datasets

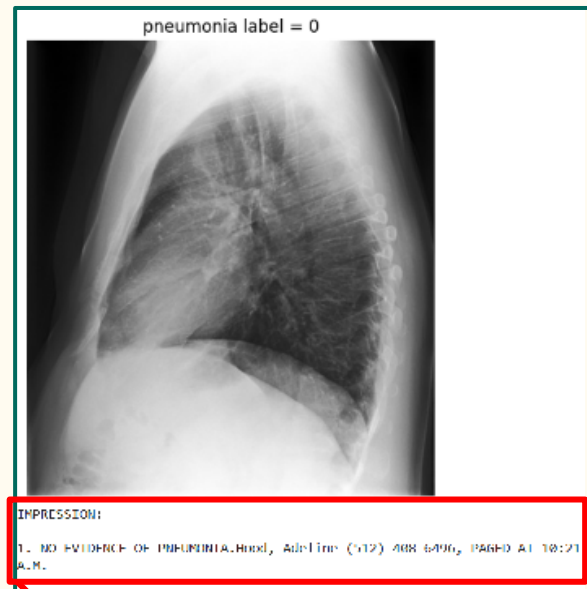
- We were given **2** datasets containing supervised (numerical) and unsupervised (categorical) data

Dataset 1



**Bounding
Boxes**

Dataset 2



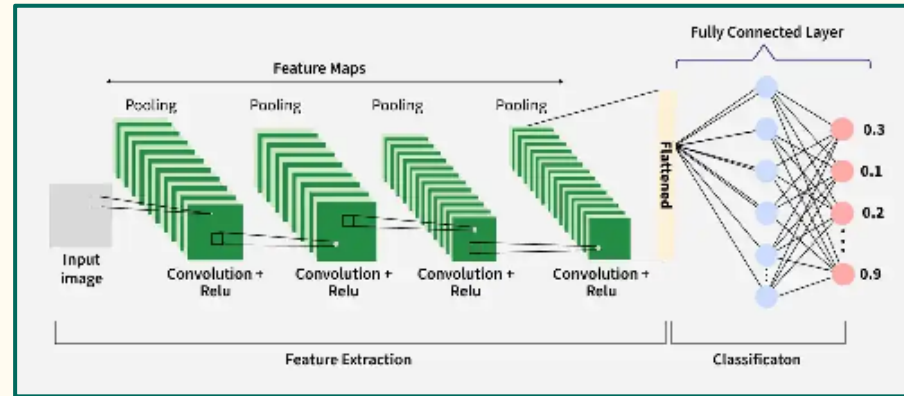
**Radiology
Reports**

2. Research Methodology

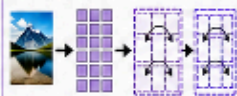
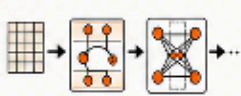
—

CNN - Why We Used It

- CNNs (Convolutional Neural Networks) are AI models that detect features in images.
- Why use a CNN?
 - CNNs use shared filters, meaning it can recognize pneumonia regardless of where it is on the chest x-ray.
 - CNNs learn useful features automatically.
 - Lightweight, efficient & learns crucial features



Models: Background Information

ResNet-18		DenseNet-121		ResNet-50		Swin V2		MaxViT	
									
	Type: CNN		Type: CNN		Type: CNN		Type: Vision Transformer		Type: Hybrid CNN + Transformer
	Key idea: residual skip connections		Key idea: dense connections (concatenate previous feature maps)		Key idea: deeper bottleneck residual blocks		Key idea: shifted-window attention		Key idea: multi-axis attention
	Depth: 18 layers		Depth: 121 layers		Depth: 50 layers		Context: local windows + broader context		Context: local + global attention
	Best for: fast, lightweight baseline		Best for: parameter efficiency + stable gradient flow		Best for: stronger CNN accuracy		Best for: scalable transformer performance		Best for: powerful all-purpose vision model
		 ~8M params vs ResNet-50 ~23M Dense routing needs efficient memory use							
ResNet-18		DenseNet-121		ResNet-50		Swin V2		MaxViT	
	Family CNN		Family CNN		Family CNN		Family Transformer		Family Hybrid
	Main strength Fast, lightweight baseline		Main strength Feature reuse, high accuracy with fewer parameters		Main strength Stronger CNN accuracy		Main strength Scalable transformer performance		Main strength Powerful all-purpose vision model
	Complexity Low		Complexity Low-Medium		Complexity Medium		Complexity Medium-High		Complexity High
ResNet + DenseNet = CNNs Swin V2 = Transformer MaxViT = Hybrid									

Ablation Studies

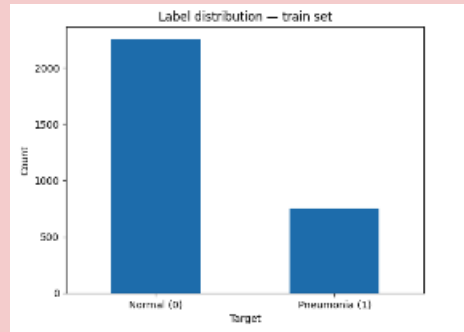
Data Augmentation

Applied random transformations to training images improve generalization and reduce overfitting
horizontal flips, small rotations, and resizing



Class Imbalance Handling

Our training dataset contains significantly more normal cases than pneumonia cases. We used class weighting to reduce bias towards the majority class



Learning Rate Optimization

We tested different learning rates to find the optimal value for stable convergence and better generalization.

Training Duration (Epochs)

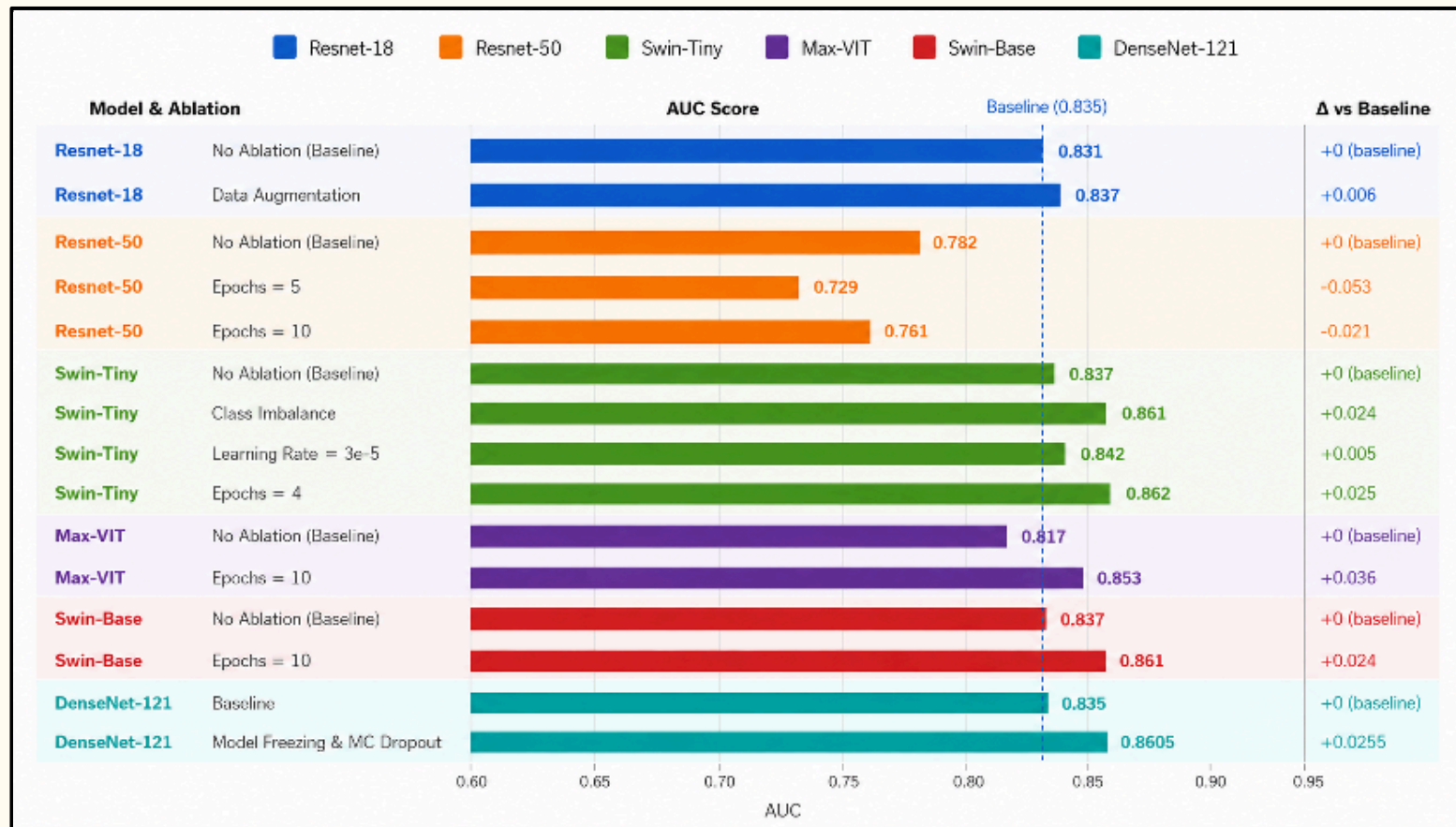
We tested different numbers of epochs to balance learning and overfitting.

Key Question: What ablation choices optimize model generalization and AUC?

3. Model Performance

—

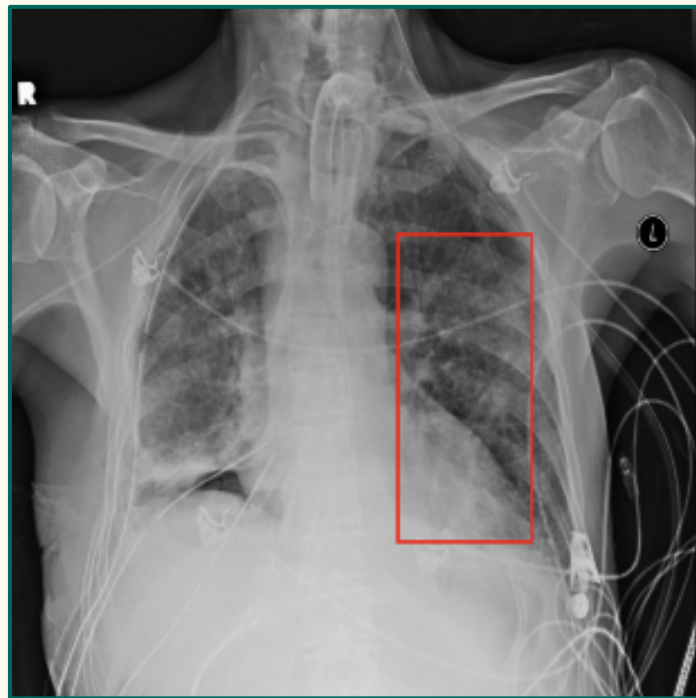
Classification Model Performance



Our best model from ablation testing was the Swin-Tiny model with no augmentations and a learning rate of $1e-4$ at 4 epochs.

Importance of Localization

- Previously, we looked at classification alone. But, when it comes to detecting pneumonia, we also want to know where it is.
- Pneumonia detection is particularly difficult because borders are often hard to define and fuzzy (as shown in the picture on the right).

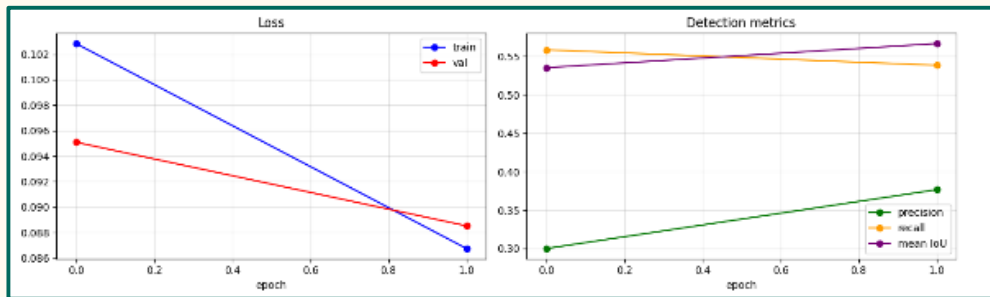


Box around a Pneumonia case on a Chest X-ray

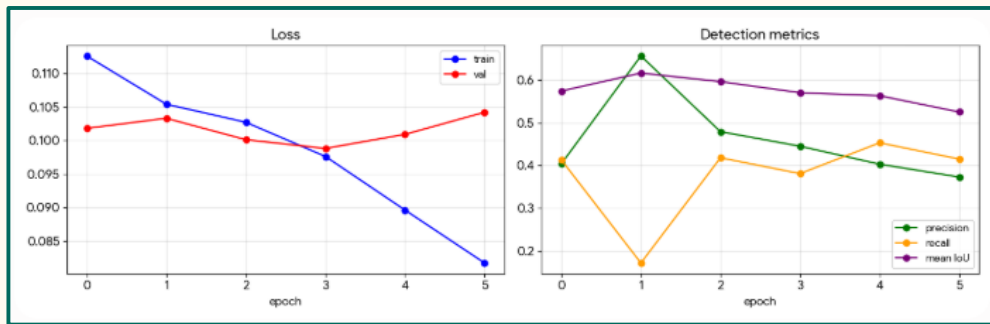
Methodology in Localization & Improvements to Model

- Backbone CNN model: Resnet-50 model, combined with fast R-CNN
- Intersection over Union (IoU): to gauge bounding box accuracy
- Initial Baseline: high F1 score of 41%
 - Decreases in training loss → no signs of overfitting
- **Methodological Improvements:**
 - Data Augmentation - random flips & color jitter to handle lighting and positional shifts
 - Learning Rate Scheduler: used to adjust learning rate across 7 epochs

Initial model with no ablations - 2nd epoch



Updated model with ablations - 7th epoch



Incorporating Text Reports

Why Multimodal AI

- Radiologists rely on both a chest x-ray and clinical notes detailing symptoms and history

👁 Vision (X-rays)

Captures patterns, opacities, and changes directly from image pixels

+

📄 Text (Reports)

Provides clinical context and detailed findings not seen in pictures

More comprehensive understanding with more context and complementary information

ResNet50 (vision: x-ray)

50-layer convolutional neural network extracts spatial features of the x-ray

+

ClinicalBert (text: report)

language model trained to understand healthcare text



Late Fusion (multimodal)

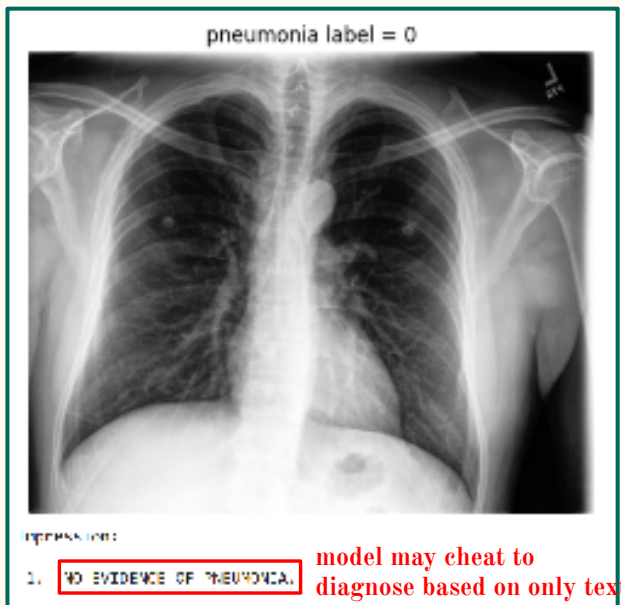
fuses vectors passed by vision and text model into one large vector



Dense Network (classification)

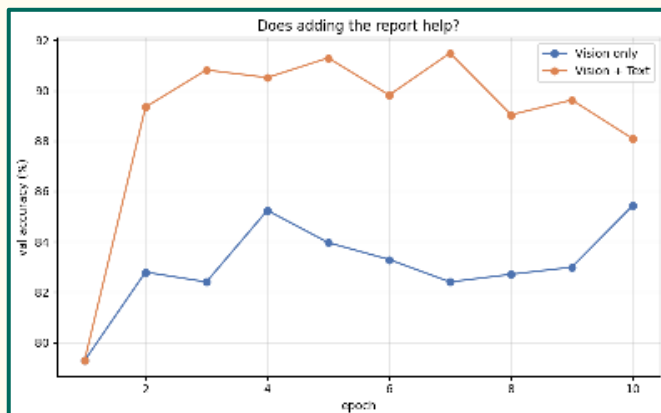
probability score of patient having pneumonia

Results



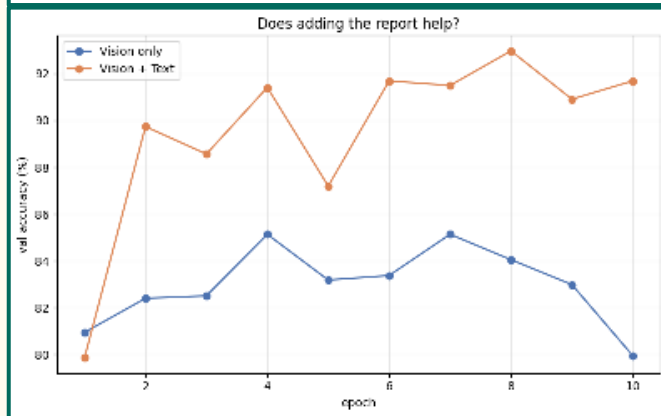
Findings:
Frontal and lateral views of the chest demonstrate a normal cardiomediastinal silhouette. The lungs are clear. The pulmonary vessels are well-defined. There is slight apical pleural thickening. There may be a small amount of right-sided pleural thickening, also related to pleural thickening. There is no evidence of pneumonia. The trachea is midline. The visualized osseous structures are unremarkable.

model must independently evaluate lung state & anomalies



vision-only: 85.4%

multimodal (I): 90.9%



vision-only: 85.1%

multimodal (F): 93%

Notebook 4 Classification Improvement

Baseline ResNet-50 vs. Dataset-2 Pretrained ResNet-50

Accuracy (0.5/best)

<u>Baseline</u>	<u>Transfer-Model</u>
0.5: 75.39%	0.5: 78.99%
Best: 79.16%	Best: 80.73%

0.5 Improvement: +3.6%
Best Improvement: +1.57%

AUROC

<u>Baseline</u>	<u>Transfer-Model</u>
0.5: 0.8504	0.8726

Improvement: +0.0222

F1 (0.5/best)

<u>Baseline</u>	<u>Transfer-Model</u>
0.5: 0.6963	0.5: 0.7181
Best: 0.7019	Best: 0.7311

0.5 Improvement: +0.0218
Best Improvement: +0.0292

4. Analysis/Broader Impacts

—

Combating Class Imbalance

Problem Statement: 3-Class Learning

Grouping normal and abnormal non-pneumonia classes into a single “normal” creates heavy dataset imbalance and confuses the model to ignore clinically meaningful differences between the two categories.

Minimizing bias due to class imbalance using augmentation and weighting

Augmentation: Applied random crop, affine, color jitter & contrast

Weighting: Applied inverse class weights to balance the dataset and prevent model shortcuts

Dropout reduces model bias and prevents overfitting

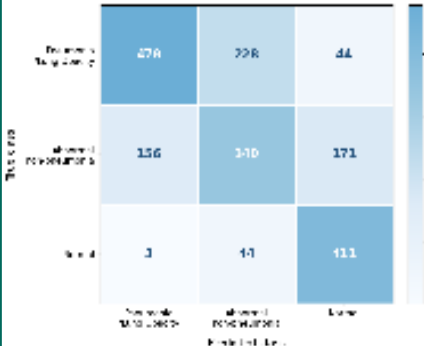
Deactivates random neurons to prevent subset bias & forces the model to generalize & learn true clinical features (not image artifacts)

Multiclass analysis helps improve accuracy by 4.82%

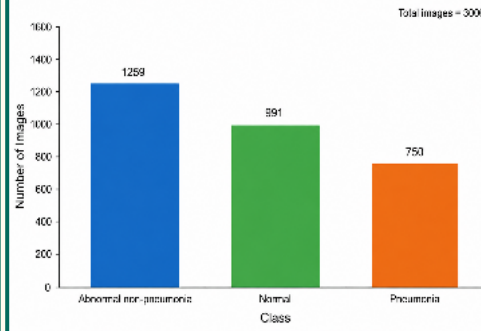
Minimizes false positives & negatives (reliable & avoids wrong predictions)

Overall, these enhancements improved AUC scores to 87%

3-Class Confusion Matrix

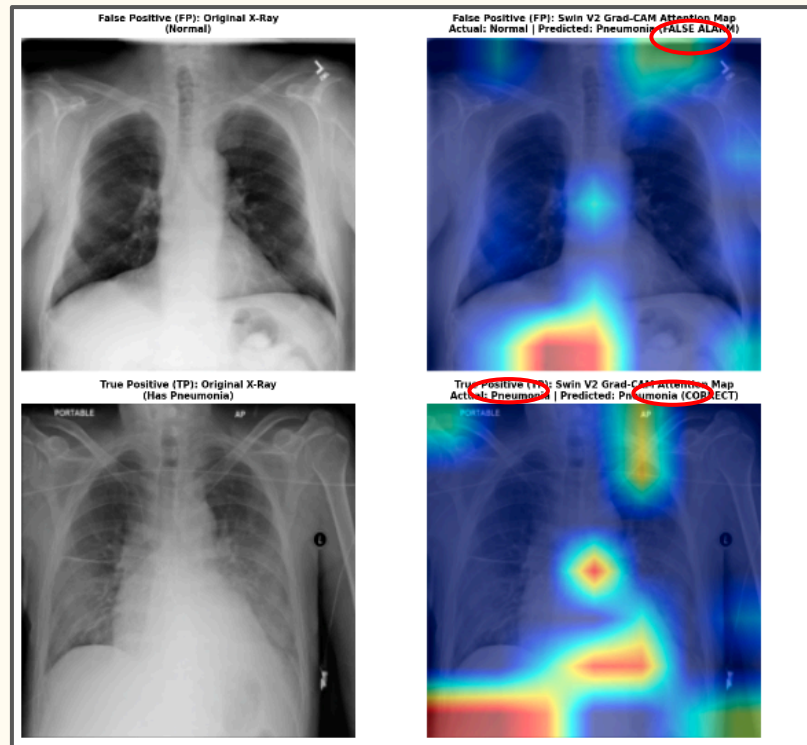


Training Set Distribution: 3-Class Formulation



Highlight Model Failures with GRAD-CAM

- The model focused mostly on the sides of the image
- The model relied on textual artifacts, such as view-position labels
- In future research, we should investigate whether cropping image edges could improve performance by reducing reliance on textual artifacts and forcing the model to focus on actual features in the X-ray.



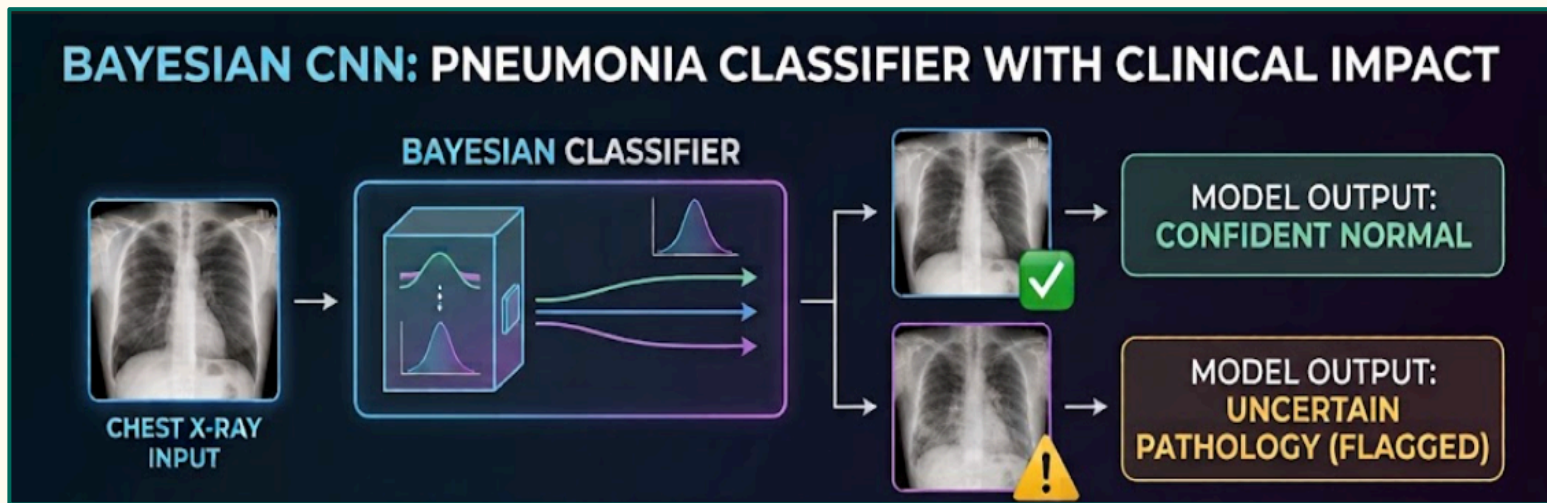
GradCAM of optimum Swin tiny model from earlier results

Pitfalls/Dead Ends

Bayesian CNN

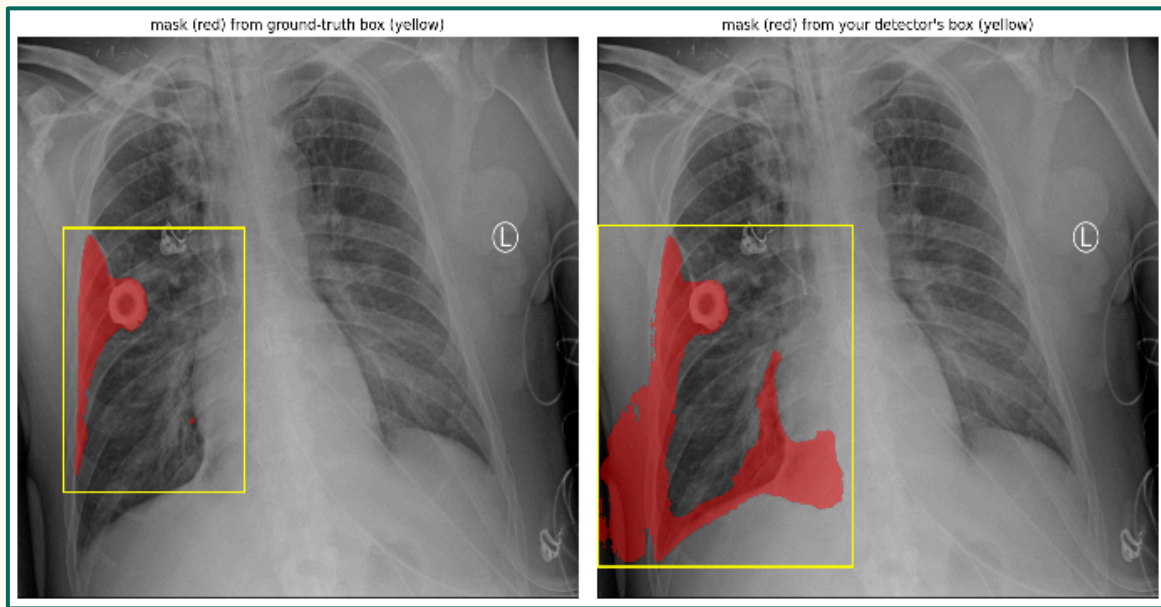
Replaces fixed network weights with probability distributions - quantifying uncertainty instead of guessing.

- Generated an average diagnosis for patient, along with a reliable confidence score - with Monte Carlo Dropout
- Automatically flags ambiguous or poor-quality scans for human review



Med SAM

- Generates segmentation masks to isolate irrelevant areas - not distracted by random artifacts
- The model finds the right spot, but has incorrect box scaling - not ready for the clinics
 - Left Image : Ground Truth
 - Right Image: Model Prediction



MedGemma - Dead End

- MedGemma is a medical AI model designed to understand and analyze healthcare text and images, such as chest X-rays.
- We used it to generate pneumonia reports which we compared to real radiologist labels.

MedGemma Report:
No Pneumonia

AI (MedGemma) report:

Pneumonia: No

Location: RLL

Key finding: Right lower lobe opacity with air bronchograms.

Confidence: High

Real radiologist impression:

1. Peripheral right lower lobe pneumonia with possible trace parapneumonic effusion

Radiologist Report:
Pneumonia

Highlights & Future Considerations

Future Considerations and Final Conclusions

By using architectures like Swin-Tiny and Fast R-CNN localization, we developed an ML pipeline that boosts diagnostic accuracy to 93%, serving as a “second pair of eyes” for radiologists

In the future, we could...

- Focus on cropping edges or applying masks to remove view-position text labels.
- External validation through Swin-Tiny and ResNet on new datasets from other hospital systems.
- Conduct pilot tests to see how much the “Clinical Assistance” pipeline actually reduces physician burnout.

Overall, models require clear uncertainty communication to prevent misdiagnosis due to class imbalances and data gaps.



PNEUMONAI

DETECT. PREDICT. BREATHE EASY.

Thank You

References

—

References

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- Stanford Center for Artificial Intelligence in Medicine and Imaging. (n.d.), *Localizing Pneumonia with Object Detection* [Google Slides]. AIMI High School Internship Research Materials. Retrieved June 21, 2026, from https://docs.google.com/presentation/d/1rYcL6Aknm6u42EA3hqoNAV7Y_n06I2sOqlOMwXHPaM/edit?usp=sharing
- Sultan, Z. K., Faridoon, F., Silkosessak, O. C.-U.-D., & Porntaveetus, T. (n.d.). *Multi-Regional deep learning models for identifying dental restorations and prosthesis in panoramic radiographs*. ResearchGate. Retrieved June 21, 2026, from https://www.researchgate.net/figure/The-Vision-Transformer-ViT-provides-a-visual-diagram-which-represents-an-image_fig5_397307390

Appendix

Appendix

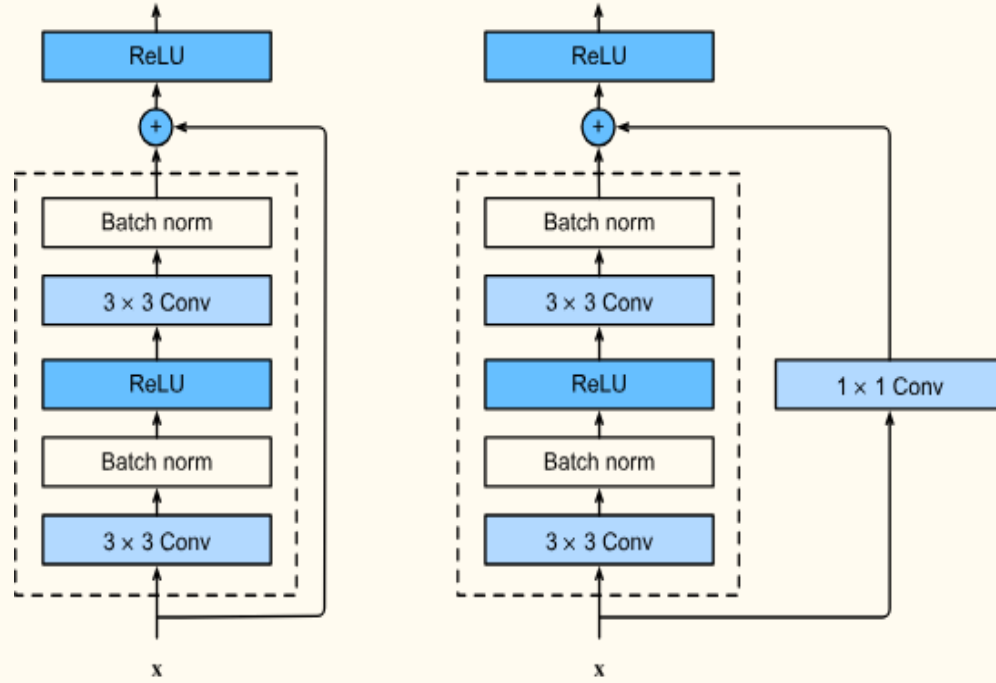
- Baseline Model Information
 - ResNet-18
 - Vision Transformers - Swin V2 & Max ViDT
 - DenseNet-121
- Baseline Model Scores
 - How R-CNN Works
 - Scores Table
- Model Deployment Impacts
 - Broad Impacts of Model Deployment

Baseline Model Information

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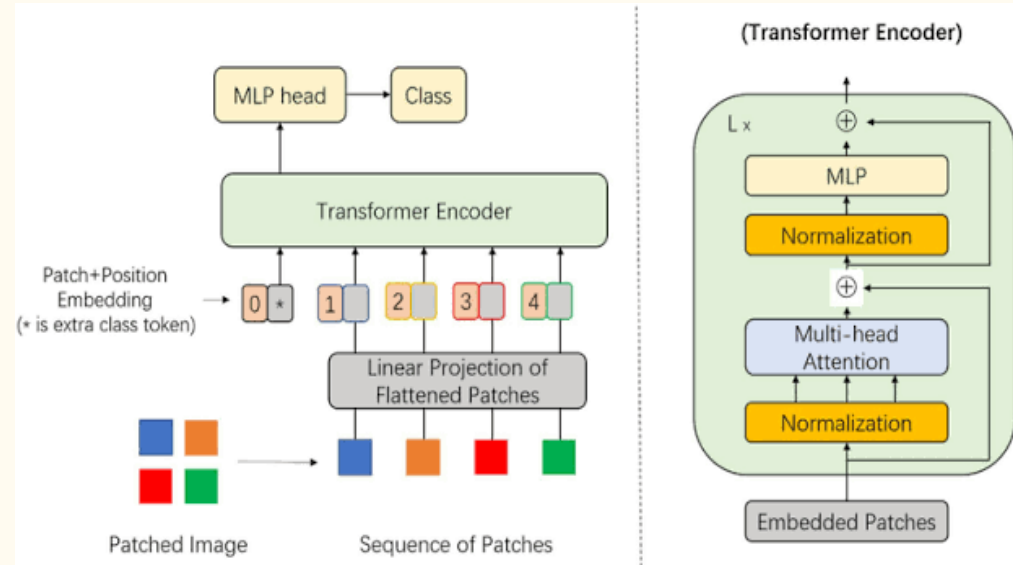
ResNet-18

- ResNet is a CNN architecture designed to train deep learning models without losing performance
- Uses “skip connections” to allow the input x to bypass the main convolutional layers $\rightarrow$ solving vanishing gradient problem



Vision Transformers - Swin V2 & MaxViT

- ViT's use the process of self-attention in order to review patches of an image and understand their relationship to each other.
- The Swin V2 transformer replaces the self-attention with a shifted window attention, dropping complexity for high res images

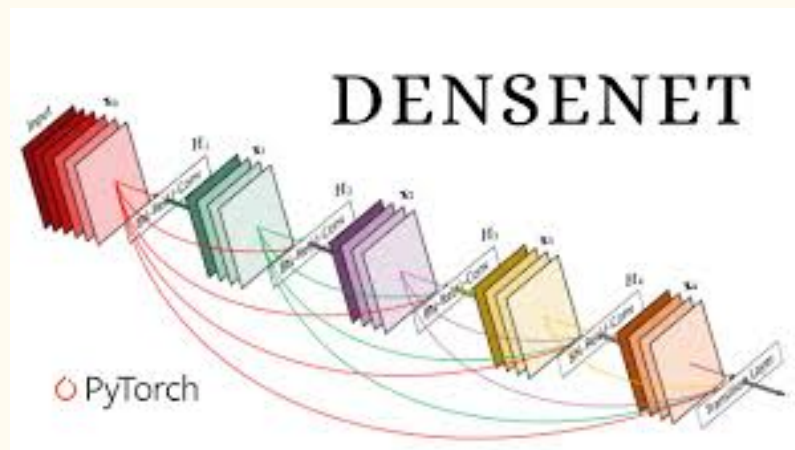


DenseNet-121

CNN with 121 layers, structured into distinct “Dense Blocks”

How it's used (Pneumonia Classification)

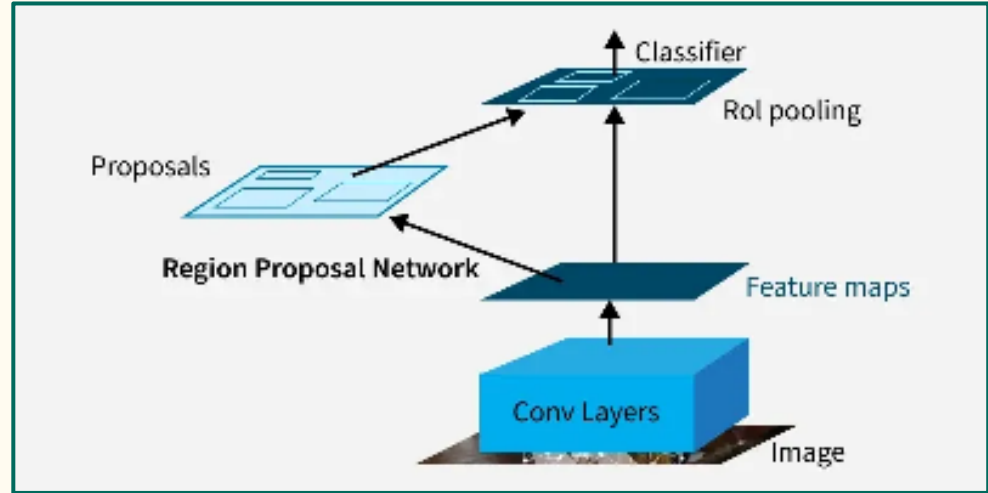
- Multimodal Backbone
 - Used to extract image features, parallel with metadata-MLP
- Two-Phase Training
 - Freezes backbone to extract med features



Baseline Model Scores

How R-CNN Works

- CNN limitation: outputs only the probability of having pneumonia
- R-CNN (Region-based CNN) generates thousands of regions within the image and runs CNN on each region.
- Fast R-CNN generates many regions, but runs CNN only once, on the entire image.



Scores Table

Model	Ablation	High Score	Δ vs Baseline
Resnet-18	No Ablation	0.831	+0 (baseline)
Resnet-18	data augmentation	0.837	+0.006
Resnet-50	no aug, lr 1e-4, 5 ep	0.729	-0.083
Resnet-50	no aug, lr 1e-4, 10 ep	0.761	-0.051
Swin-Tiny	No Ablation	0.837	+0 (baseline)
Swin-Tiny	Class Imbalance	0.861	+0.03
Swin-Tiny	Learning Rate = 3e-5	0.842	+0.011
Max-ViT	no aug, lr 1e-4, 10 ep	0.853	+0.02
Swin-Base	no aug, lr 1e-4, 10 ep	0.861	+0.03
Swin-Tiny	No aug, lr 1e-4, 4 ep	0.862	+0.025
Swin-Tiny	Aug, early stop, class imbalance (weighted), lr=5e-5, split into 3 classes	0.862	+0.031

Model Deployment Impacts

Broad Impacts of Model Deployment

- Without sufficient training, even with high accuracy, a model can showcase false negatives and fail to detect pneumonia
- Class imbalance in datasets creates fairness issues. Certain patient groups such as patients from different hospitals with different imaging equipment may not be well represented or the model may perform poorly if it is not generalized sufficiently.
- A given model should not be used as a standalone diagnostic system unless it has been externally validated. It should support clinicians and not replace them. The system should clearly communicate uncertainty, especially when confidence is low.